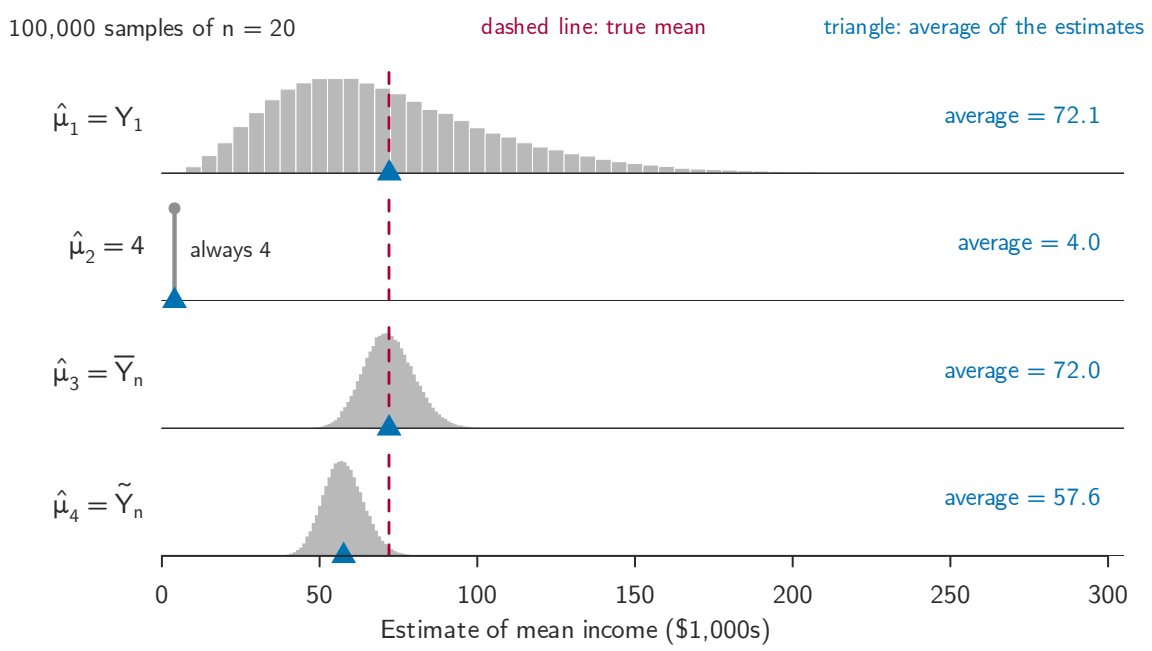
100,000 samples of n = 20
dashed line: true mean
triangle: average of the estimates
$\hat{\mu}_1 = Y_1$
average = 72.1
$\hat{\mu}_2 = 4$
always 4
average = 4.0
$\hat{\mu}_3 = \overline{Y}_n$
average = 72.0
$\hat{\mu}_4 = \tilde{Y}_n$
average = 57.6
0
50
100
150
200
250
300
Estimate of mean income ($1,000s)